

42

sin, cos, tan を含む関数の極限

(例1) 次の極限を調べよ。

point

○ をそろえてっ→まあわせ

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \left(\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1 \right)$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \quad \left(\lim_{x \rightarrow 0} \frac{x^2}{1 - \cos x} = 2 \right)$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \quad \left(\lim_{x \rightarrow 0} \frac{x}{\tan x} = 1 \right)$$

$$(1) \quad \lim_{x \rightarrow 0} \frac{\sin 2x}{x} \quad \leftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \cdot 2$$

$$= 1 \cdot 2$$

$$= 2 //$$

$$(2) \quad \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos 2x} \quad \leftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{x^2}{1 - \cos 2x} \cdot \frac{1}{4}$$

$$= 2 \cdot \frac{1}{4}$$

$$= \frac{1}{2} //$$

$$(3) \quad \lim_{x \rightarrow 0} \frac{\tan^2 x}{x} \quad \leftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{(\tan x)^2}{x}$$

$$= 1^2 \cdot 0$$

$$= 0 //$$

$$(4) \quad \lim_{x \rightarrow 0} \frac{\sin 3x}{\tan 2x} \quad \leftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{2x}{\tan 2x} \cdot \frac{3}{2}$$

$$= 1 \cdot 1 \cdot \frac{3}{2}$$

$$= \frac{3}{2} //$$

$$(5) \quad \lim_{x \rightarrow 0} \frac{\sin x}{\cos x - 1} \quad \leftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \left(-\frac{x^2}{1 - \cos x} \cdot \frac{\sin x}{x} \cdot \frac{1}{x} \right) \quad \leftarrow -2 \cdot 1 \cdot \frac{1}{0}$$

そこで

$$\lim_{x \rightarrow 0} \frac{\sin x}{\cos x - 1} = -\infty$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{\cos x - 1} = \infty$$

であるから、極限はない //

$$(6) \quad \lim_{x \rightarrow 0} \frac{\sin(1 - \cos x)}{x^2} \quad \leftarrow \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin(1 - \cos x)}{1 - \cos x} \cdot \frac{1 - \cos x}{x^2}$$

$$= 1 \cdot \frac{1}{2}$$

$$= \frac{1}{2} //$$

(例2) 次の極限を求めよ。

$$(1) \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} \quad (2) \quad \lim_{x \rightarrow 0} x \sin \frac{1}{x^2} \quad (3) \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 2x}{\cos x}$$

$$(4) \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{4 \cos^2 x - 3}{1 - 2 \sin x} \quad (5) \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{\sin^3 x - 1}$$

point

 $x \rightarrow a$ ならば $t \rightarrow 0$ となるように置き換え

※ 式変形 → 約分も有効

(1) $t = x - \frac{\pi}{2}$ とおくと

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} = \lim_{t \rightarrow 0} \frac{\cos(t + \frac{\pi}{2})}{t}$$

$$= \lim_{t \rightarrow 0} \frac{-\sin t}{t}$$

$$= -1 //$$

(2) $t = \frac{1}{x^2}$ とおくと

$$\lim_{x \rightarrow 0} x \sin \frac{1}{x^2} = \lim_{t \rightarrow 0} \frac{1}{t} \sin t^2$$

$$= \lim_{t \rightarrow 0} \frac{\sin t^2}{t^2} \cdot t \quad \text{訂正}$$

$$= 1 \cdot 0$$

$$= 0 //$$

(3) $t = x - \frac{\pi}{2}$ とおくと

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 2x}{\cos x} = \lim_{t \rightarrow 0} \frac{\sin 2(t + \frac{\pi}{2})}{\cos(t + \frac{\pi}{2})}$$

$$= \lim_{t \rightarrow 0} \frac{-\sin 2t}{-\sin t}$$

$$= \lim_{t \rightarrow 0} \left(\frac{\sin 2t}{2t} \cdot \frac{t}{\sin t} \cdot 2 \right)$$

$$= 1 \cdot 1 \cdot 2$$

$$= 2 //$$

(別解)

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 2x}{\cos x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{2 \sin x \cos x}{\cos x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} 2 \sin x \quad \leftarrow 2 \cdot 1$$

$$= 2 //$$

$$(4) \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{4 \cos^2 x - 3}{1 - 2 \sin x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{4(1 - \sin^2 x) - 3}{1 - 2 \sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - 4 \sin^2 x}{1 - 2 \sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 + 2 \sin x)(1 - 2 \sin x)}{1 - 2 \sin x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} (1 + 2 \sin x) \quad \leftarrow 1 + 2 \cdot \frac{1}{2}$$

$$= 2 //$$

$$(5) \quad \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{\sin^3 x - 1} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 x}{(\sin x - 1)(\sin^2 x + \sin x + 1)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 + \sin x)(1 - \sin x)}{(\sin x - 1)(\sin^2 x + \sin x + 1)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(-\frac{1 + \sin x}{\sin^2 x + \sin x + 1} \right) \quad \leftarrow -\frac{1+1}{1^2+1+1}$$

$$= -\frac{2}{3} //$$